

Sandesh Thapa

CONTACT INFORMATION

Robotics & Controls Research Engineer
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RESEARCH INTERESTS

Safety critical control and motion planning for autonomous and robotics systems.

EDUCATION

Oklahoma State University, Stillwater, OK August 2016 – December 2018
Master of Science, Mechanical and Aerospace Engineering (GPA 3.9/4.0)
Concentration: Control Theory & Robotics
Advisor: Dr. He Bai; **Co-Advisor:** Dr. Jose A. Acosta; **Committee:** Dr. Rushikesh Kamalapurkar
Thesis: *Cooperative Aerial Manipulation with Force Control and Attitude Stabilization*
Award: Best Graduate Research, MAE Research Symposium, 2018

McNeese State University, Lake Charles, LA August 2011 – May 2015
Bachelor of Science, Mechanical Engineering (Major GPA 3.8/4.0)
Minor: Mathematics

PUBLICATIONS

- J1. **Thapa S.**, Self R., Bai H., Kamalapurkar R. *Cooperative Manipulation of an Unknown Payload with Concurrent Mass and Drag Force Estimation*. IEEE Control Systems Letters (L-CSS), with CDC presentation option, 2019.
- J2. **Thapa S.**, Bai H., Acosta J.A. *Cooperative Aerial Manipulation with Decentralized Adaptive Force-Consensus Control*. Journal of Intelligent and Robotic Systems (JINT), January 2020.
- C1. **Thapa S.**, Bai H., Acosta J.A. *Cooperative Aerial Load Transport with Attitude Stabilization*. American Control Conference (ACC), 2018.
- C2. **Thapa S.**, Bai H., Acosta J.A. *Force Control in Cooperative Aerial Manipulation*. IEEE International Conference on Unmanned Aircraft Systems (ICUAS), 2018.
- C3. **Thapa S.**, Bai H., Acosta J.A. *Cooperative Aerial Load Transport with Force Control*. IFAC Workshop on Networked and Autonomous Air and Space Systems (NAASS), 2018.

PUBLICATIONS IN PREPARATION

- J1. **Thapa S.** *A Safety-Aware State-Machine Wrapper for Legacy PID Loops with a Head-to-Head Comparison to Control Barrier Function Safety Filters*. In preparation for IEEE Robotics and Automation Letters (RA-L).
- J2. **Thapa S.** *Clothoid-Based Path Planning with a Closed-Form Single-Trial Feasibility Predicate and Nonlinear Lateral Tracking for Low-Speed Automated Parking*. In preparation for IEEE Intelligent Vehicles Symposium (IV).
- T1. **Thapa S.** *A Comparative Tutorial on Autonomous Quadrotor Trajectory Tracking Control: PID, State-Dependent LQR, Geometric SE(3), and Nonlinear Adaptive Control*. In preparation for arXiv release.

RESEARCH EXPERIENCE

University of Texas at Arlington – ETAIC Lab, Arlington, TX
Visiting Researcher, Department of Electrical Engineering September 2026 – Present

- Research host: Dr. H. Eric Tseng (NAE Member, Distinguished University Professor).
- Design, implementation, and experimental validation of safety-critical control algorithms for autonomous systems and robotics.

- Focus: compositional safety for vehicle dynamics – composing actuator-level guard architectures (input boundedness, integrator boundedness, bumpless transfer) with state-space safety filters (CBF-QP, reachability-based filters) for end-to-end safety under embedded deployment constraints.
- Target output: peer-reviewed publication in IEEE Transactions on Control Systems Technology or IEEE Transactions on Intelligent Vehicles.

Independent Research, Jersey City, NJ

Autonomous Systems Research

February 2026 – August 2026

- Design, MATLAB/Simulink implementation, and real-time validation of quadrotor trajectory tracking controllers (PID, state-dependent LQR, geometric SE(3), nonlinear adaptive) in Flightmare, Gazebo, and ROS.
- Formulation of a closed-form single-trial feasibility predicate for clothoid-based planning in low-speed automated parking.
- Preparation of three papers for submission to IEEE RA-L, IEEE IV, and arXiv.

Amogy Inc., Brooklyn, NY

Tech Lead and Senior Controls Research Engineer

August 2022 – February 2026

- Designed and deployed a safety-aware discrete-event state-machine wrapper for legacy PID control loops, establishing analytical properties of input boundedness, integrator boundedness, and bumpless mode transfer. Deployed across multiple safety-critical autonomous systems with ISO 26262 functional-safety considerations. (Basis of IEEE RA-L submission in preparation.)
- Head-to-head evaluation of the state-machine wrapper against control-barrier-function safety filters on systems where a clean analytical CBF construction is unavailable due to lack of model or invariant set.
- Research on nonlinear, hybrid, and predictive control frameworks for safety-critical autonomous operation; implementation on automotive-grade real-time hardware (dSPACE, VCU).
- **Tools:** C/C++, Python, MATLAB/Simulink, Stateflow, dSPACE, CAN, TCP/IP.

Ford Motor Company – Research and Advanced Engineering, Dearborn, MI

Research Engineer, Autonomous Driving (Planning and Controls)

January 2021 – June 2022

- Advisor: Dr. H. Eric Tseng (NAE Member).
- Developed a continuous-curvature (clothoid-based) path planner and a nonlinear rear-wheel feedback lateral controller for autonomous parallel parking and auto-hitch (SAE L2/L3 features).
- Prototyped autonomous-vehicle trajectory tracking controllers including pure-pursuit, LQR, PD, backstepping, and nonlinear feedback control; validated in simulation and on-vehicle in real time.
- Designed state machines and control-barrier-function based safety controllers for autonomous function transitions and fault handling.
- Developed V2V networking and real-time TCP/UDP interfaces for cooperative localization and collision avoidance.
- **Tools:** Simulink, Stateflow, ROS, C/C++, TCP/IP, dSPACE, CANape, CANalyzer.

The Drone Racing League, New York, NY

Senior Controls Engineer, Autonomy

July 2019 – January 2021

- Led research, design, and implementation of trajectory tracking, attitude stabilization, and trajectory generation algorithms for autonomous quadrotor UAVs.
- Implemented state-dependent LQR/LQG, nonlinear PID, model reference adaptive control (MRAC), geometric SE(3) control, and model predictive control (MPC).

- Built ROS/Gazebo and PX4 SITL simulation environments and conducted real flight tests on quadrotor hardware.
- Implemented C++ real-time solvers for the algebraic Riccati equation using Schur decomposition, eigenvalue decomposition, and Newton iteration.
- **Tools:** C/C++, ROS, PX4, Gazebo, Eigen, CMake, Python, MATLAB/Simulink.

Oklahoma State University, Stillwater, OK

Graduate Research Assistant, CoRAL Lab

August 2016 – June 2019

- Research on cooperative control and estimation for multi-robot systems, focused on multi-UAV cooperative manipulation of a shared unknown payload.
- Developed decentralized adaptive controllers for cooperative aerial payload transport with no direct inter-robot communication, handling unknown payload mass and drag.
- Developed a concurrent-learning adaptive control algorithm for real-time parameter estimation, relaxing the persistent-excitation requirement while preserving parameter convergence.
- Five peer-reviewed publications: IEEE L-CSS (with CDC presentation), JINT, ACC, ICUAS, IFAC NAASS.

Graduate Teaching Assistant

August 2016 – December 2017, August 2018 – December 2018

- Elementary Dynamics (ENSC 2123): discussion sessions, partial lectures, grading.
- Mechatronics Design (MAE 4733): lab instruction and grading (28 students).

SELECTED
RESEARCH
PROJECTS

Compositional Safety for Autonomous Systems (Current)

- Formalization of a discrete-event guard architecture as a safety layer over legacy PID control loops with provable input-boundedness, integrator-boundedness, and bumpless-transfer properties.
- Composition of the guard architecture with CBF-QP and reachability-based state-space safety filters for end-to-end safety on aerial and ground vehicle platforms.

Autonomous Vehicle Planning and Control (Ford R&A)

- Clothoid-based continuous-curvature path planner and nonlinear rear-wheel lateral feedback controller for parallel parking and auto-hitch.
- State machines and CBF-based safety certificates for autonomous function transitions.
- Trajectory tracking benchmarks: pure-pursuit, LQR, PD, backstepping. dSPACE real-time hardware validation.

Autonomous Quadrotor UAV Control and Planning (DRL, MS)

- Cascaded PID, state-dependent LQR, nonlinear Lyapunov-based, sliding-mode, backstepping, geometric SE(3), and Cartesian impedance controllers for quadrotor trajectory tracking.
- Differential-flatness-based minimum-snap trajectory generation.
- End-to-end autopilot integrating state estimation, planning, and control; MATLAB/Simulink, ROS/Gazebo, PX4 SITL simulation environments; real quadrotor flight tests.

Decentralized Cooperative Aerial Manipulation (MS Thesis)

- Decentralized adaptive force-consensus controllers enabling a multi-UAV team to jointly transport a payload without direct inter-robot communication under unknown payload mass and external disturbances.
- Stability analysis via Lyapunov methods; validated in simulation and in physical multi-UAV flight tests.

Concurrent Learning-Based Adaptive Estimation (MS)

- Concurrent-learning adaptive controller for real-time estimation of unknown payload mass and drag during cooperative transport; guaranteed parameter convergence with relaxed excitation requirements.
- Accurate force regulation and synchronized multi-agent motion demonstrated in high-fidelity simulation.

TECHNICAL SKILLS **Programming:** C/C++, Python, MATLAB/Simulink, Stateflow
Robotics and Simulation: ROS, Gazebo, PX4-SITL, Flightmare, Crazyflie UAVs, Eigen
Real-Time Hardware: dSPACE, MicroAutoBox II, VCU, Raspberry Pi, Jetson TX2/Xavier
Automotive and Communication: CAN, CANape, CANalyzer, CarSim, TCP/UDP Networking, Git

CERTIFICATIONS AND NANODEGREES **Certifications:** Functional Safety Certification, 2024
Udacity Nanodegrees: Intro to Self-Driving Cars, 2021; Flying Car and Autonomous Flight Engineer, in progress

AWARDS AND FELLOWSHIPS

- Ross Fellowship, Purdue Aerospace Engineering PhD, Fall 2021 (admission deferred and later declined).
- Dean's Fellowship, CU Boulder ECE PhD, Fall 2021 (declined).
- Full GRA-supported PhD admissions at Georgia Tech (ECE), UIUC (Aerospace), University of Maryland (Aerospace), and University of Delaware (Mechanical), Fall 2021 (declined).
- Best Graduate Research, MAE Research Symposium, Oklahoma State University, 2018.
- Winner, Nautilus Engineering Design Challenge, 2014–2015.
- First Place, McNeese Undergraduate Research Competition, 2015.
- H.R. Smith Engineering Scholarship.

GRADUATE COURSEWORK	Linear Systems	Nonlinear System Analysis and Control
	Robotics: Kinematics, Dynamics, and Control	Digital Control Systems
	Atmospheric Flight Control	Guidance and Control of Aerospace Vehicles
	Stochastic Systems	Intro to Modern Analysis
	Optimal Control (Audit)	Adaptive Control (Audit)

PEER REVIEWER

- American Control Conference (ACC): 2017–2022, 2025.
- IEEE Conference on Decision and Control (CDC): 2018, 2019, 2021, 2022.
- IEEE International Conference on Unmanned Aircraft Systems (ICUAS): 2018, 2021.
- IEEE International Conference on Robotics and Automation (ICRA): 2018, 2020, 2021.
- Journal of Intelligent and Robotic Systems (JINT): 2020.

REFERENCES Available upon request.